MAST20009 Vector Calculus

Problem Sheet 6: Integral Theorems

Worked Solutions for selected questions

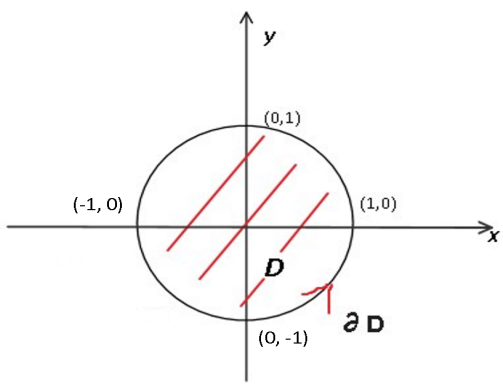
Q94. (a)

Solution: Given $D : x^2 + y^2 \leq 1,\ \boldsymbol{F}(x,\ y) = x\mathbf{i} + y\mathbf{j}$.

To verify:

$$\iint_D \boldsymbol{\nabla} \cdot \boldsymbol{F}\, dx\, dy = \int_{\partial D} \boldsymbol{F} \cdot \hat{\boldsymbol{n}}\, ds$$

where $\hat{\boldsymbol{n}}$ is outward normal in the xy-plane to ∂D.



$$\boldsymbol{\nabla} \cdot \boldsymbol{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) = 1 + 1 = 2$$

$$\text{LHS} = \iint_D \boldsymbol{\nabla} \cdot \boldsymbol{F}\, dx\, dy = \iint_D 2\, dx\, dy = 2 \text{ area of } D$$
$$= 2\pi, \qquad \text{as } D \text{ is a disk of radius } 1\,.$$

Now

$$\hat{\boldsymbol{n}} = \hat{\boldsymbol{r}} = \frac{(x, y)}{\sqrt{x^2 + y^2}} = (x, y) \qquad \text{on } \partial D$$

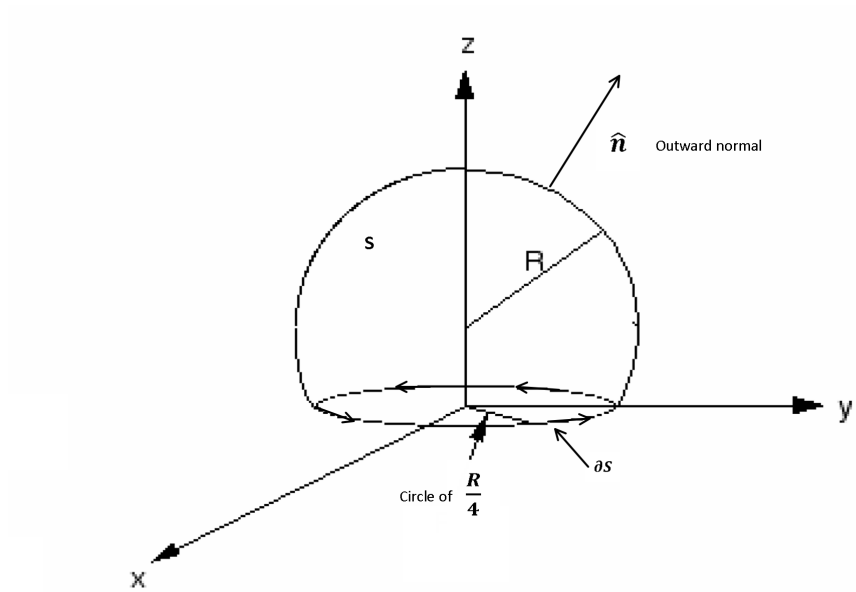
$$\Rightarrow \boldsymbol{F} \cdot \hat{\boldsymbol{n}} = (x, y) \cdot (x, y) = x^2 + y^2 = 1 \qquad \text{on } \partial D.$$

So

$$\text{RHS} = \int_{\partial D} \boldsymbol{F} \cdot \hat{\boldsymbol{n}} ds = \int_{\partial D} 1 ds = \text{ length of } \partial D$$
$$= 2\pi \qquad \text{since } \partial D \text{ is a circle of radius 1.}$$

Q98. c
Solution:

$$\boldsymbol{F}(x,y,z) = \boldsymbol{\nabla} \times (-y, x, 0), \qquad R = 5$$



$$\begin{aligned}\text{Volume flow rate (flux)} &= \iint_S \boldsymbol{F} \cdot \hat{\boldsymbol{n}} dS = \iint_S (\boldsymbol{\nabla} \times (-y, x, 0)) \cdot d\boldsymbol{S} \\ &= \int_{\partial D} (-y, x, 0) \cdot d\boldsymbol{c} \qquad \text{by Stokes' theorem}\end{aligned}$$

Parametrise ∂D by $\mathbf{c}(t) = (x(t), y(t), z(t)),\ x(t) = \dfrac{R}{4}\cos t,\ y(t) = \dfrac{R}{4}\sin t,\ z(t) = 0,\ 0 \le t \le 2\pi$

$$\Rightarrow x'(t) = -\tfrac{R}{4}\sin t,\ y'(t) = \tfrac{R}{4}\cos t,\ z'(t) = 0 \Rightarrow \mathbf{c}'(t) = \ (-\tfrac{R}{4}\sin t,\ \tfrac{R}{4}\cos t, 0).$$

So,

$$\begin{aligned}\text{flux} &= \int_0^{2\pi} \left(-\frac{R}{4}\sin t,\ \frac{R}{4}\cos t, 0\right) \cdot \left(-\frac{R}{4}\sin t,\ \frac{R}{4}\cos t, 0\right) dt \\ &= \int_0^{2\pi} \frac{R^2}{16}\sin^2 t + \frac{R^2}{16}\cos^2 t dt = \int_0^{2\pi} \frac{R^2}{16} dt = \frac{\pi R^2}{8} = \frac{25\pi}{8}\,.\end{aligned}$$

Q101. a
Solution:

$$\boldsymbol{F}(x,y,z) = e^x \sin y\, \mathbf{i} + e^x \cos y\, \mathbf{j} + z^2 \mathbf{k}$$

$$\boldsymbol{c}(t) = \left(\sqrt{t},\ t^3,\ e^{\sqrt{t}}\right), \qquad 0 \le t \le 1$$

$$\boldsymbol{\nabla} \times \boldsymbol{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^x \sin y & e^x \cos y & z^2 \end{vmatrix} = (0-0)\mathbf{i} - (0-0)\mathbf{j} + (e^x \cos y - e^x \cos y)\mathbf{k} = 0\,.$$

Since $\boldsymbol{F}$ is C^1 in $\mathbb{R}^3$ and $\boldsymbol{\nabla} \times \boldsymbol{F} = 0$, $\boldsymbol{F}$ is conservative field. Hence $\boldsymbol{F} = \boldsymbol{\nabla}\phi$ for some ϕ.

So

$$\left(\frac{\partial \phi}{\partial x},\ \frac{\partial \phi}{\partial y}\ ,\ \frac{\partial \phi}{\partial z}\right) = (e^x \sin y,\ e^x \cos y,\ z^2)$$

Equate the components

$$\frac{\partial \phi}{\partial x} = e^x \sin y \Rightarrow \phi = e^x \sin y + c_1(y,z) \tag{1}$$

$$\frac{\partial \phi}{\partial y} = e^x \cos y \Rightarrow \phi = e^x \sin y + c_2(x,z) \tag{2}$$

$$\frac{\partial \phi}{\partial z} = z^2 \Rightarrow \phi = \frac{1}{3}z^3 + c_3(x,y) \tag{3}$$

Equations (1-3) give

$$\phi(x,y,z) = e^x \sin y + \frac{1}{3}z^3 + c \quad \text{where } c \in \mathbb{R}\,.$$

Hence

$$\int_C \boldsymbol{F} \cdot d\boldsymbol{c} = \phi(\boldsymbol{c}(1)) - \phi(\boldsymbol{c}(0)) = \phi(1,1,\ e) - \phi(0,0,1) = e\sin(1) + \frac{1}{3}e^3 - \frac{1}{3}\,.$$